RELATIVE EQUIVALENCE KERNELS DESCEND THE LIMIT TOWER

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Perfect descent. perfect-lemma-descend-perfect, perfect-lemma-flat-proper-perfect-direct-image-general (cross-chapter, named). Descend K and K^prime; preserve perfectness of the convolution.
- Smooth diagonal perfectness. more-morphisms-lemma-smooth-diagonal-perfect, more-morphisms-lemma-perfect-proper-perfect-direct-image (cross-chapter, named). The diagonal kernel at level i is perfect.

Source reference. Original definition 0G0I: sources/stacks-equiv/pinned-original-source.tex, source lines 3792–3792. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0K: sources/stacks-equiv/pinned-original-source.tex, source lines 3838–3838. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B6G: focused-seeds/geometry/S9-eqv-descend-rek/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FDP: focused-seeds/geometry/S9-eqv-descend-rek/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09RF: focused-seeds/geometry/S9-eqv-descend-rek/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B91: focused-seeds/geometry/S9-eqv-descend-rek/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $S = \lim_{i \in I} S_i$ be a limit of a directed system of schemes with affine transition morphisms $g_{i'i} : S_{i'} \to S_i$. We assume that S_i is quasi-compact and quasi-separated for all $i \in I$. Let $0 \in I$. Let $X_0 \to S_0$ and $Y_0 \to S_0$ be smooth proper morphisms. We set $X_i = S_i \times_{S_0} X_0$ for $i \geq 0$ and $X = S \times_{S_0} X_0$ and similarly for Y_0. If K is the Fourier-Mukai kernel of a relative equivalence from X to Y over S then for some $i \geq 0$ there exists a Fourier-Mukai kernel of a relative equivalence from X_i to Y_i over S_i.*

Proof. Let $K' \in D_{perf}(\mathcal{O}_{Y \times_S X})$ be the object assumed to exist in Definition 0G0I. Since $X \times_S Y = \lim X_i \times_{S_i} Y_i$ there exists an i and objects K_i and K'_i in $D_{perf}(\mathcal{O}_{Y_i \times_{S_i} X_i})$ whose pullbacks to $Y \times_S X$ give K and K'. See Derived Categories of Schemes, Lemma 09RF. By Derived Categories of Schemes, Lemma 0B91 the object

$$R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^* K_i \otimes^{\mathbf{L}}_{\mathcal{O}_{X_i \times_{S_i} Y_i \times_{S_i} X_i}} L\mathrm{pr}_{23}^* K'_i)$$

is perfect and its pullback to $X \times_S X$ is equal to

$$R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X \times_S Y \times_S X}} L\mathrm{pr}_{23}^* K') \cong \Delta_{X/S,*}\mathcal{O}_X$$

See proof of Lemma 0G0K. On the other hand, since $X_i \to S$ is smooth and separated the object

$$\Delta_{i,*}\mathcal{O}_{X_i}$$

of $D(\mathcal{O}_{X_i \times_{S_i} X_i})$ is also perfect (by More on Morphisms, Lemmas 0FDP and 0B6G) and its pullback to $X \times_S X$ is equal to

$$\Delta_{X/S,*}\mathcal{O}_X$$

See proof of Lemma 0G0K. Thus by Derived Categories of Schemes, Lemma 09RF after increasing i we may assume that

$$\Delta_{i,*}\mathcal{O}_{X_i} \cong R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^* K_i \otimes^{\mathbf{L}}_{\mathcal{O}_{X_i \times_{S_i} Y_i \times_{S_i} X_i}} L\mathrm{pr}_{23}^* K'_i)$$

as desired. The same works for the roles of K and K' reversed. □